Reza Ghorbani Saber

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Profile

- Operations Research specialist in production planning and scheduling: combinatorial, decomposition, and stochastic optimization.
- Translates industrial planning problems into decision-support tools, from analysis and design to implementation and validation.
- Industrial production planning experience, working with shop-floor teams, contractors, and executive management.
- Programming in Python and C/C++; Gurobi, IBM ILOG CPLEX, CP Optimizer, Git, CMake, Docker, and HPC.

Professional Experiences

Ghent University & Flanders Make, Ghent, Belgium — Feb 2022-Present
PhD Researcher & Algorithm Developer

- Designed and implemented C++ and Python scheduling engines for flexible job shops with machines, operator skills, and shift rules.
- Formulated deterministic and stochastic optimization models addressing operational uncertainty, workforce side constraints, and dual-resource machine/operator allocation.
- Implemented Logic-Based Benders Decomposition (MIP, CP) and tabu search; cut the average optimality gap from 37% to 6% on 262 instances.
- Collaborated with industrial partners in the S4FMS project, building APIs to integrate optimization engines into partner architectures and industrial testbeds, where the operator-timetabling scheduler is being tested.
- Presented results at international conferences (EURO, ORBEL, INCOM, APMS) and published instances and results as open data.

Taha Ghaleb Toos company (TGT Co), Mashhad, Iran — May 2020- Oct 2021
Project Planning and Control Specialist

- Directed production planning and resource scheduling across Machine, CNC, Laser Cladding, and Assembly shops for 20+ industrial R&D projects.
- Built an automated MS Access/Excel database and reporting pipeline, delivering real-time shop-floor tracking, capacity visibility, and financial progress dashboards to executive management.
- Relieved operational bottlenecks and contractor delays by translating machine constraints and job priorities into actionable production timetables.

Education

Ghent University, Ghent, Belgium — 2022-Present
Doctor of philosophy in Industrial Engineering and Operations Research

Ferdowsi University of Mashhad, Mashhad, Iran — 2016-2018
MSc in Industrial Engineering-Systems Optimization

Ferdowsi University of Mashhad, Mashhad, Iran — 2012-2016
BSc in Industrial Engineering

Publications

Ghorbani Saber, R., Leyman, P., & Aghezzaf, E. H. (2026). Decision support for the integrated flexible job shop scheduling and operator timetabling problem. *International Journal of Production Research*, 1-28. https://doi.org/10.1080/00207543.2026.2642881

Ghorbani Saber, R., Leyman, P., & Aghezzaf, E. H. (under review). A Benders decomposition-based hybrid approach to the dual resource constrained flexible job shop scheduling problem. *Computers & Operations Research*.

Ghorbani Saber, R., & Ranjbar, M. (2022). Minimizing the total tardiness and the total carbon emissions in the permutation flow shop scheduling problem. *Computers & Operations Research*, *138*, 105604. https://doi.org/10.1016/j.cor.2021.105604

Ranjbar, M., & **Ghorbani Saber, R**. (2021). A variable neighborhood search algorithm for transshipment scheduling of multi products at a single station. *Applied Soft Computing*, *98*, 106736. https://doi.org/10.1016/j.asoc.2020.106736

Languages

English	Full professional proficiency (C1, IELTS 7.5)
Dutch	Intermediate (B1)
Persian	Native